

PORTFOLIO REPORT

Linear Infrastructure Digitization: Legacy Raster-to-Vector Conversion

Project Name: Kent County Interurban Transit Network Modernization

Role: Lead GIS Specialist

Date: May 2026

1. Executive Summary

A non-spatial raster asset (JPEG) depicting the Kent County interurban and suburban transit network was converted into a fully attributed, topologically coherent vector geodatabase. The transformation enables network-level spatial analysis, asset management, and future AI-driven transit optimization. Manual digitization and rigorous topology enforcement were executed in QGIS 3.x, producing a schema-compliant GeoPackage ready for enterprise integration.

2. Technical Methodology

Phase I: Spatial Realignment (Georeferencing)

The raster was imported into QGIS and georeferenced using the Georeferencer GDAL plugin. A Point-to-Point method matched identifiable intersections (e.g., Division Ave & Fulton St) with real-world coordinates on an OpenStreetMap base layer.

Phase II: Manual Vectorization

A new GeoPackage container was generated with LineString geometry. High-fidelity manual tracing was performed at a scale of 1:2,000, snapping to the georeferenced raster centerlines.

Phase III: Topology Validation

Vertex and segment snapping tolerances were enforced to guarantee coincident geometry at transit hubs. A connectivity check confirmed that all interurban lines (Silver, Laker) intersected the central downtown hub without errors.

3. Database Schema & Specifications

Field Name	Data Type	Description	Example
Route_ID	Text (String)	Unique segment identifier	SL-100 (Silver Line)
Frequency	Text (String)	Temporal service classification	Peak, Off-peak
Asset_Type	Text (String)	Infrastructure tier	Interurban, Suburban

4. Technical Specifications

Component	Specification
Software	QGIS 3.34 LTR
Primary CRS	EPSG:3857 – WGS 84 / Pseudo-Mercator
Output Formats	GeoPackage (.gpkg), ESRI Shapefile (.shp)
Topology Tolerance	10 meters (snapping)

"The deliverable is an analysis-ready network dataset that enables service area generation, total track mileage calculations, and AI/ML model readiness for urban digital twin initiatives."